JEE APEX FTS – JEE ADV. FULL SYLLABUS TEST-08
2023 [P1]
KEY SHEET

MATHEMATICS

1	**ABCD**	2	**ABC**	3	**CD**	4	**B**	5	**C**	6	**A**
7	**A**	8	**2**	9	**4**	10	**100**	11	**1**	12	**40**
13	**8**	14	**A**	15	**B**	16	**D**	17	**C**		

PHYSICS

18	**AD**	19	**ABCD**	20	**BCD**	21	**C**	22	**C**	23	**C**
24	**C**	25	**3**	26	**2**	27	**2**	28	**10**	29	**2**
30	**4**	31	**D**	32	**C**	33	**A**	34	**C**		

CHEMISTRY

35	**BC**	36	**AD**	37	**AC**	38	**A**	39	**B**	40	**A**
41	**B**	42	**5**	43	**6**	44	**6**	45	**4**	46	**4**
47	**121**	48	**A**	49	**A**	50	**C**	51	**D**		

SOLUTIONS
MATHEMATICS

1. $X = {}^{20}C_3 = 1140$

$Y = \dfrac{{}^{20}C_1 \cdot {}^{16}C_2}{3} = 800$

$Z = {}^{20}C_3 - 10 \times {}^{18}C_1 = 1140 - 180 = 960$

$Y \cap Z = Y - 10 \times 14 = 800 - 140 = 660$

2. $\because a(1-e) = 5 \Rightarrow a = 20 \qquad \therefore b^2 = 400\left(1 - \dfrac{9}{16}\right) = 175$

$\because P_1P_2 = b^2 \ \therefore P_1 \times 5 = 175 \Rightarrow P_1 = 35$

Also centre of ellipse is (-11, -8)

3. Using characteristic equation $\lambda_1 = 2, \lambda_2 + \lambda_3 = 3$

$|A| = 2$

We get $A^3 - 5A^2 + 7A - 2I = 0$

(A) $\dfrac{1}{81}|\text{adj } 2A|\left(\text{adj}3A^{-1}\right)|A^{-1}| = \dfrac{1}{81} \times |2A|^2 \left|3A^{-1}\right|^2 \dfrac{1}{|A|} = \dfrac{1}{81} \times 2^6 \times 3^6 \times \dfrac{1}{2} = 32 \times 9 = 288$

$(B)\left(A^{-1}\right)^2 = \dfrac{1}{2}\left(A - 5I + 7A^{-1}\right)$

$(C)\left(A^5 - 2A^4\right) = |A|^4 |A - 2I| = 0$

As 2 is an eigen value D) $2A^{-1} = 2I + A^2 - 5A$

$A^{-1} = \dfrac{\left(A^2 - 5A + 7I\right)}{2}$

4. Put x = 0, $f(0).e^{(f(0))} = 0 \Rightarrow f(0) = 0$

$\Rightarrow$ f is bijective so $f(x) > 0 \forall x \in (0, \infty)$

$f(x)\, e^{f(x)} = x \qquad \ln f(x) + f(x) = \ln x$

$\dfrac{f(x)}{\ln x} = 1 - \dfrac{\ln f(x)}{\ln x} \ldots\ldots(1)$

We know that $e^x > x$

$\dfrac{\sqrt{x}}{x} > \dfrac{f(x)}{x} \Rightarrow 0 < \dfrac{f(x)}{x} < \dfrac{1}{\sqrt{x}}$

$\lim\limits_{x\to\infty} \dfrac{f(x)}{x} = 0 \Rightarrow$ growth of f < growth of 'x'

$\Rightarrow$ growth of ln f(x) < growth of lnx $\Rightarrow \lim\limits_{x\to\infty} \dfrac{\ln f(x)}{\ln x} = 0$

Putting this value in equation (1)

$\lim\limits_{x\to\infty} \dfrac{f(x)}{\ln x} = 1$

5. $x + xy + xyz = 1 \Rightarrow x(1+y) = 1 - xyz \ldots\ldots(1)$

$y + yz + xyz = 2 \Rightarrow y(1+z) = 2 - xyz \ldots\ldots(2)$

$z + zx + xyz = 4 \Rightarrow z(1+x) = 4 - xyz \ldots\ldots(3)$

Multiply all above equations

$xyz(1+x)(1+y)(1+z) = (1-xyz)(2-xyz)(4-xyz)$

let $xyz = p, (1+x)(1+y)(1+z) = q$

$pq = (1-p)(2-p)(4-p)$

$q = 1 + x(1+y) + y(1+z) + z(1+x) + p$

$q = 1 + 1 - p + 2 - p + 4 - p + p$

$q = 8 - 2p$

$p(8-2p) = (1-p)(2-p)(4-p)$

$(4-p)(p^2 - 5p + 2) = 0$

$p = 4, \frac{5+\sqrt{17}}{2}. \frac{5-\sqrt{17}}{2}$

Largest value of xyz = $\frac{5+\sqrt{17}}{2}$

$a = 5, b = 17, c = 2$

$a + b + c = 24$

6. $\frac{1}{\sqrt{k}+\sqrt{k+1}} = \sqrt{k+1} - \sqrt{k} \Rightarrow \sum_{k=1}^{80}\left(\sqrt{k+1} - \sqrt{k}\right) < \sum_{k=1}^{80}\frac{1}{2\sqrt{k}} \Rightarrow \sum_{k=1}^{80}\frac{1}{\sqrt{k}} > 16$

Similarly $\sum_{k=2}^{80}\left(\sqrt{k} - \sqrt{k-1}\right) > \sum_{k=1}^{80}\frac{1}{2\sqrt{k}} \Rightarrow S = \left(1 + 2\sqrt{80} - 2\right) < 17$

7. $\int\left(\sin(n+1)x + x\right)\sin^n x\, dx$

$\int \sin(n+1)x.\ \sin^n x \cos x\, dx +$

$\int \cos(n+1)x.\ (\sin x)^{(n+1)}\, dx$

$\frac{(\sin x)^{(n+1)}}{(n+1)}\sin(n+1)x - \int \cos(n+1)x.\ (\sin x)^{n+1} +$

$\int \cos\left((n+1)x\right).\ (\sin x)^{n+1} x$

$F_n(\pi) = 0 \Rightarrow c = 0$

Hence $F_n(x) = \frac{(\sin x)^{(n+1)}}{(n+1)}\sin(n+1)x$

$F_{16}\left(\frac{3\pi}{2}\right) = -\frac{1}{17}\sin\frac{51\pi}{2} = a$

$a = \frac{1}{17}$

8. $x-4=t+1\Rightarrow x-5=t\therefore \sum_{r=0}^{3n}a_r(1+t)^r=\sum_{r=0}^{3n}A_rt^r$

$a_0(1+t)^0+a_1(1+t)^1+.....+a_{2n}(1+t)^{2n}+$

$a_{2n+1}(1+t)^{2n+1}+.....+a_{3n}(1+t)^{3n}=A_0t^0+A_1t^1+.....+A_{2n}t^{2n}+.....+A_{3n}t^{3n}$

Compare coefficient of t^{2n} on both sides $A_{2n}=a_{2n}.\ {}^{2n}C_{2n}+a_{2n+1}\ {}^{2n+1}C_{2n}+...+a_{3n}\ {}^{3n}C_{2n}$

$={}^{2n}C_{2n}+{}^{2n+1}C_{2n}+{}^{2n+2}C_{2n}+.....+{}^{3n}C_{2n}=\left({}^{2n+1}C_{2n+1}+{}^{2n+1}C_{2n}\right)+.....+{}^{3n}C_{2n}$

$A_{2n}={}^{3n+1}C_{2n+1}$

similiarly $B_{2n}={}^{3n+1}C_{2n+1}$

$D_{2n}={}^{3n+1}C_{2n+1}\quad\therefore\dfrac{A_{2n}+D_{2n}}{B_{2n}}=2$

9. $\sqrt{3}(\bar{a}.\bar{b})+\left|\bar{b}-(\bar{a}.\bar{b})\bar{a}\right|=1+|\bar{b}|^2$

$\sqrt{3}(\bar{a}.\bar{b})+\left|\bar{a}\times(\bar{b}\times\bar{a})\right|=1+|\bar{b}|^2$

$\sqrt{3}(\bar{a}.\bar{b})+|\bar{a}||\bar{b}\times\bar{a}|\sin 90^0=1+|\bar{b}|^2$

$\sqrt{3}(\bar{a}.\bar{b})+|\bar{a}\times\bar{b}|=1+|\bar{b}|^2$

$\sqrt{3}|\bar{b}|\cos\theta+|\bar{b}|\sin\theta=1+|\bar{b}|^2$

$\dfrac{\sqrt{3}\cos\theta+\sin\theta}{\le 2}=\dfrac{|b|+\dfrac{1}{|b|}}{\ge 2}\therefore\theta=\dfrac{\pi}{6}\qquad\therefore\operatorname{cosec}^2\theta=4$

10. $\left[\{x\}\right]=0$

$f(x)=\left(3-x^7\right)^{\frac{1}{7}}$

$f(f(x))=\left(3-\left(3-x^7\right)^7\right)^{\frac{1}{7}}=x$

$f(f(x))=x\Rightarrow f^{-1}(x)=f(x)\qquad\therefore f^{-1}(50)=f(50)$ and $f(f(100))=100$

$\therefore f^{-1}(50)-f(50)+f(f(100))=100$

11. Let $S_1=\alpha+\beta+\gamma$ and $S_2=\alpha\beta+\beta\gamma+\gamma\alpha$ and $S_3=\alpha\beta\gamma$

$\therefore S_1=6$ and $(\alpha+1)(\beta+1)(\gamma+1)=33\Rightarrow S_3+S_1+S_2=32$

$\Rightarrow S_2+S_3=26\ \ .....(1)$

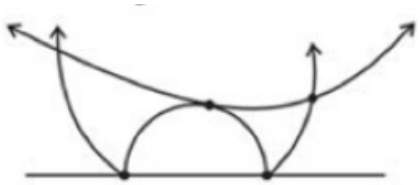
Also, $\alpha^3+\beta^3+\gamma^3=87\qquad\Rightarrow S_1\left(S_1^2=3S_2\right)=3S_3=87$

$\Rightarrow 216-18S_2+S_3=87\qquad\Rightarrow 6S_2-S_3=43.....(2)$

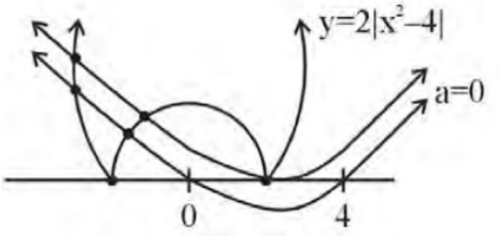
Adding (1) and (2) $\Rightarrow 7S_2=69\Rightarrow S_2=\dfrac{69}{7}$ and $S_3=\dfrac{113}{7}$

$\therefore\dfrac{1}{\alpha}+\dfrac{1}{\beta}+\dfrac{1}{\gamma}=\dfrac{S_2}{S_3}=\dfrac{69}{113}$

12. Case – I : When For 3 solution $y = x^2 - 4x + a$ touches
$y = 8 - 2x^2$ in $[-2, 2] \Rightarrow x^2 - 4x + a = 8 - 2x^2$
$\Rightarrow 3x^2 - 4x + a - 8 = 0$ has exactly one root $16 - 12(a - 8) = 0$
$\Rightarrow 4 - 3a + 24 = 0 \Rightarrow 28 = 3a \qquad \Rightarrow a = \frac{28}{3}$

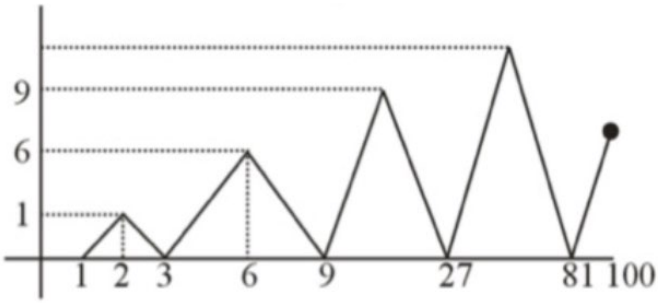


(note : L.R of $y = x^2 - 4x + a >$ L.R of $y = 2x^2 - 8$)



When $y = x^2 - 4x + a$ passes through (2, 0) a = 4

13. $f(x) = \begin{cases} x - 1, & x \in [1, 2) \\ 3 - x, & x \in [2, 3] \end{cases}$



f(x) is non-differentiable at x = 2, 6, 12, 20, 30, 42, 56, 72

14. $(P)\ f'(x) = 0 \Rightarrow 2x \sin\frac{\pi}{x} + x^2 \cos\frac{\pi}{x}\left(-\frac{\pi}{x^2}\right) = 0 \Rightarrow \tan\frac{\pi}{x} = \frac{\pi}{2x}$
$\Rightarrow$ put $\frac{\pi}{x} = \theta \therefore \theta \in (\pi, 21\pi)$ as $x \in \left(\frac{1}{21}, 1\right)$
Now number of solution of $\tan\theta = \frac{\theta}{2}$ in $(\pi, 21\pi)$ is 20
(Q) let a and b roots of f(x) = 0
Now, f(f(1)) = f(f(-2)) = 0 but $f(1) \neq f(-2) \Rightarrow f(1) = a\ f(-2) = b$
$\Rightarrow 1 + p + q + 3 = a,\ 4 - 2p + q + 3 = b$
$\Rightarrow$ adding above two equations $2q - p + 11 = a + b \Rightarrow q = -\frac{11}{2}$
also, $3p - 3 = a - b \Rightarrow 9p^2 - 18p + 9 = p^2 - 4(q + 3) \Rightarrow 8p^2 - 18p - 1 = 0$
$\Rightarrow g(2) = 4 + 2(8p^2 - 18p) - q + 2 = \frac{27}{2}$

(R) $f(x)=\left(x^9-9x^8\right)+\left(6x^8-54x^7\right)+\left(x^4-9x^3\right)+\left(x^3-9x^2\right)+\left(x^2-9x\right)+x+9$

$\Rightarrow f(9)=18$

(S) $L=e^{\lim\limits_{x\to 0}\left(\tan^{-1}\frac{1}{x^2}+x^2\sin\frac{1}{x^4}\right)}=e^{\frac{\pi}{2}}$

15. (I) Line L = y = x $\Rightarrow$ image of P(2, 4) in L will be Q(4, 2)

Circles $S:(x-4)^2+(y-2)^2=16$

$\Rightarrow x^2+y^2-8x-4y+4=0$

Circle M touches y = x at (1, 1)

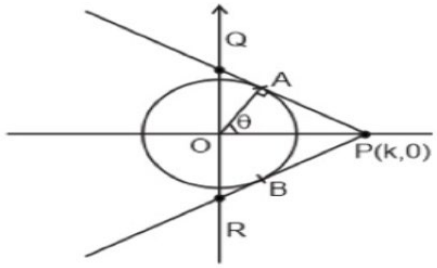
$M:(x-1)^2+(y-1)^2+\lambda(x-y)=0$

Now radical axis of circles, S, M : S – M = 0

$M: x^2+y^2+7x-11y+2=0$

$\text{radius}=\sqrt{\frac{49}{4}+\frac{121}{4}-2}=\frac{\sqrt{162}}{2}$

(II)



Here $OP = K,\ OA=\sqrt{50}$

$K.\cos\theta=\sqrt{50}$

$K=\sqrt{50}\sec\theta$

also $\angle OQP=\theta \quad =\frac{100}{\sin 2\theta}\Rightarrow \text{min area}=100$

when $\theta=45^0 \quad \Rightarrow K=\sqrt{50}\times\sec 45^0=\sqrt{50},\sqrt{2}=10$

(III) $2x^2+xy+y^2=5\rightarrow$ centre $(0, 0)$

put $x = r,\cos\theta,\ y = r\sin\theta \Rightarrow r^2=\frac{5}{2\cos^2\theta+\sin\theta.\cos\theta+\sin^2\theta}$

$=\frac{10}{2+2\cos^2\theta+2\sin\theta.\cos\theta}$

For length of major axis ‘r’ is maximum

$r^2_{max}=\frac{10}{3-\sqrt{2}}\Rightarrow a=3,\ b=\sqrt{2} \qquad \Rightarrow a^2.\ b=9.\sqrt{2}=\sqrt{162}$

(IV) tangent to $\frac{x^2}{25}-\frac{y^2}{24}=1$ at $\left(5\sqrt{2},\ 2\sqrt{6}\right)$ will be:

$\frac{x.5\sqrt{2}}{25}-\frac{y.2\sqrt{6}}{24}=1$

Intersecting tangent at vertices $x=\pm 5$ at points A, B

$$\Rightarrow \pm\sqrt{2}-\frac{y}{2\sqrt{6}}=1 \Rightarrow \pm\sqrt{2}-\frac{y}{2\sqrt{6}}=1$$

Now segment of any tangent between tangent at vertices subtends 90^0 at foci, circle drawn through these 4 points will have A, B as end points of diameter.

$$\Rightarrow K=\frac{2\sqrt{6}\left(\sqrt{2}-1\right)-2\sqrt{6}\left(\sqrt{2}+1\right)}{2}=-2\sqrt{6} \Rightarrow \left|2\sqrt{3}.K\right|=12.\sqrt{2}$$

16. $n(s)={}^{30}C_4$

(I) $P=1-P(\text{are no are} \geq 15) \qquad =-1\frac{{}^{16}C_4}{{}^{30}C_4}=\frac{731}{783}$

(II) $P=\frac{{}^{19}C_1\times 1\times {}^{10}C_2}{{}^{30}C_4}=\frac{19}{609}$

(III) $P=P(4 \text{ even})+P(2E, 2O)+P(4O)=\frac{{}^{15}C_4+{}^{15}C_2\times {}^{15}C_2+{}^{15}C_4}{{}^{30}C_4}=\frac{2951}{5481}$

(IV) P = 1 – P(pqrs is odd) $=1-\frac{{}^{15}C_4}{{}^{30}C_4}=\frac{248}{261}$

17. (I) $\int_0^1 f(x)-f^{-1}(x)\,dx=0$

(II) $I=\frac{\int_0^{51}([x]+25)dx}{\int_0^{51}\{x\}\,dx}=\frac{(0+1+.....+50)+25\times 51}{51\times\frac{1}{2}}$

$$I=\frac{\frac{50\times 51}{2}+25\times 51}{\frac{51}{2}}=50\times 2=100$$

(III) $L=\lim_{n\to\infty}\frac{1}{n}\sum_{r=1}^{n}\sin^2\frac{r\pi}{2n}$

$L=\int_0^1\sin^2\frac{x\pi}{2}dx \qquad L=\int_0^1\sin^2\left(\frac{\pi}{2}(1-x)\right)dx$

$2L=\int_0^1 1\,dx$

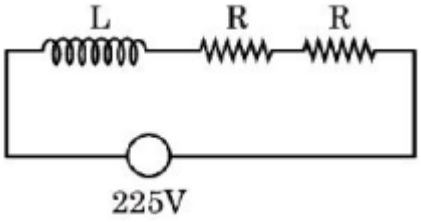
$L=\frac{1}{2}$

(IV) $J=\int_0^\infty xe^{-2x}\,dx=\frac{xe^{-2x}}{-2}\Big|_0^\infty-\int_0^\infty\frac{e^{-2x}}{-2}\,dx=(0,-0)+\frac{1}{2}\frac{e^{-2x}}{-2}\Big|_0^\infty$

PHYSICS

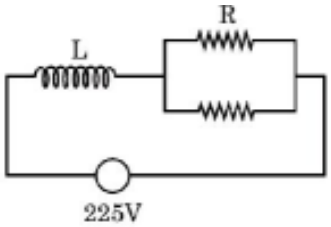
18. $\frac{2\pi}{\lambda}=K=\frac{2\pi}{0.8}\ \lambda=0.8m\ \frac{\lambda}{u}=20cm\ l+e=100cm=\frac{5\lambda}{u}$

Hence second over tone closed organ pipe



19. $P_1=V_{rms}i_{rms}\cos\phi\quad P_1=\frac{V_{rms}^2}{Z}.\frac{R_{eq}}{Z}=\frac{(225)^2.2R}{\sqrt{X_L^2+(2R)^2}}$

$$810=\frac{(225)^2.2R}{\sqrt{X_L^2+4R^2}}.....(i)$$



$P_2=V_{rms}i_{rms}\cos\phi\quad P_2=\frac{V_{rms}^2}{Z}=\frac{R_{eq}}{Z}=\frac{(225)^2(R/2)}{\sqrt{X_L^2+\left(\frac{R}{2}\right)^2}}$

$$506.25=\frac{(225)^2R}{2\sqrt{X_L^2+\frac{R^2}{4}}}.....(ii)$$

From (i) and (ii), we get $R=20\Omega, X_L=30\Omega$

and $\tan\phi_2=\frac{X_L}{R_{eq}}=\frac{X_L}{(R/2)}=\frac{2X_L}{R}\ \tan\phi_2=\frac{60}{20}=3\ \phi_2=\tan^{-1}(3)$

20. $V=\frac{kQ}{R}=\frac{9}{10}=0.9V$

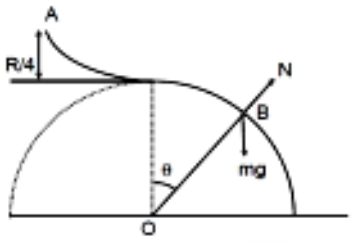
$V_{S_1}=\frac{hc}{l\lambda}-\phi=\frac{1240}{250}-4=0.96>0.9\Rightarrow$ same will escape

$V_{S_2}=\frac{1240}{300}-4<0\Rightarrow$ no electrons come out.

$V_{S_3}=\frac{1240}{300}-4=0.13V\Rightarrow$ Electrons come out and reenter

$V_{S_4}=\frac{1240}{155}-4=4eV\Rightarrow$ Many will escape.

21. Conserving energy between points A and B, $mg\left[\frac{R}{4}+R(1-\cos\theta)\right]=\frac{1}{2}mv^2$

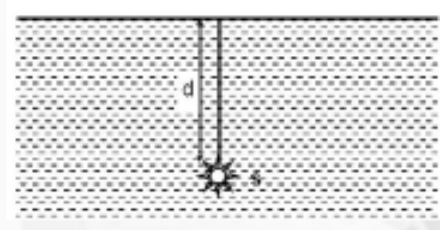


Also at point B, N = 0 so, $mg\cos\theta = \frac{mv^2}{R}$

$\frac{1}{4} + (1-\cos\theta) = \frac{1}{2}\cos\theta \frac{5}{4} = \frac{3}{2}\cos\theta \ \cos\theta = \frac{5}{6}$

22. $C = C_V + \frac{dW}{ndT}, \int dW = \int_{T_0}^{3T_0} \alpha \ TdT - \int_{T_0}^{3T_0} \frac{5R}{2} dT$ for one mole $W = (4\alpha T_0 - 5R)T_0$

23.



$f = \frac{2\pi(1-\cos\theta_c)}{4\pi} f = \frac{2\pi}{4\pi}\left[1-\sqrt{1-\frac{1}{\mu^2}}\right] = \frac{1}{2}\left[1-\sqrt{1-\frac{1}{\mu^2}}\right]$

24. $E2\ \pi r = \frac{dB}{dt}(B\pi r^2) E = \frac{r}{2}\frac{dB}{dt}$

$d\tau = (2\pi r\ dr)\sigma.\ E.r\ \tau = \int d\tau = \int \frac{r}{2}\frac{dB}{dt}(2\pi r\ dt)\sigma.r$

25. Apply torque equations due to pseudo force and weight. $mg\frac{l}{2} + mgl = ma\frac{l}{2}$ $a = 3g$

26. Current in wire = $\frac{3}{30} = 0.1$ A $V = IR = 2V$

Maximum voltage can read = 2V $\lambda = \frac{V}{l} = \frac{2}{10\times10^3} = 0.2$ mV / mm

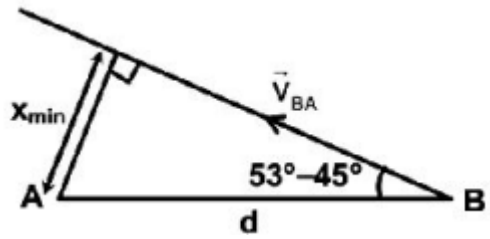
Least account of scale = 1 mm accuracy = 0.2 mV

27. $\alpha = \frac{R_2 l_1}{R_1 - R_2} = \frac{(200)(33)-(100)(67)}{100-200} = 1cm$

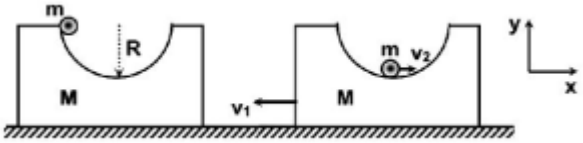
$\beta = \frac{R_1 l_1 - R_2 l_2}{R_1 - R_2} - 100 = \frac{(33)(100)-(200)(67)}{100-200} - 100 = 1cm$

28. $\frac{U_e}{U_m} = \frac{1}{8}\mu_0\varepsilon_0\omega^2 R^2 \quad = 5\times10^{-15}$

29. $x_{min} = d\sin(53^0 - 45^0) = \frac{d}{5\sqrt{2}} = 2$



30. Momentum conservation along x-axis $0 = mv_2 - Mv_1$

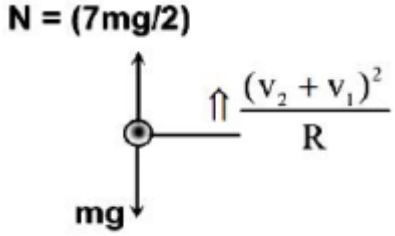


Work energy theorem $mgR = \frac{1}{2}mv_2^2 + \frac{1}{2}Mv_1^2$

from (i) and $(ii) \Rightarrow 2mgR = mv_2^2 + \frac{m^2}{M}v_2^2$

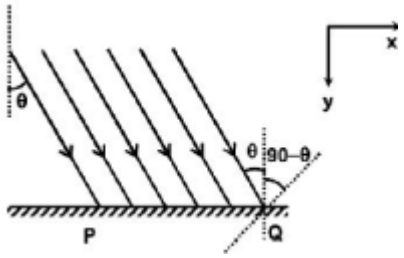
$\sqrt{\frac{2gRM}{M+m}} = v_2$ and $v_1 = \frac{m}{M}\sqrt{\frac{2gRM}{M+m}}$

Making FBD of particle with respect wedge at lowest point. $\frac{7mg}{2} - mg = \frac{m(v_1+v_2)^2}{R}$



Substituting the values, we get : $\frac{M}{m} = 4$

31. Intensity is given by $I = \frac{N\left(\frac{hC}{\lambda}\right)}{(A\cos\theta)\Delta t}$, where N – total number of photons and $\lambda \rightarrow$ Wavelength of photons



(A) $F_x = \frac{\Delta P_x}{\Delta t} = \frac{IA\cos\theta.\sin\theta}{C}$

$F_y = \frac{\Delta P_y}{\Delta t} = \frac{IA\cos\theta.\cos\theta}{C}$ $p_r = \frac{F_y}{A} = \frac{I\cos^2\theta}{C}$

(B) $F_x = 0\ F_y = \dfrac{2IA\cos^2\theta}{C}\ p_r = \dfrac{2IA\cos^2\theta}{C}$

(C) and (D)

$F_x = \dfrac{IA\cos^2\theta}{C}(1-r)\ F_y = \dfrac{IA\cos^2\theta}{C}(1+r)\ p_r \dfrac{I\cos^2\theta}{C}(1+r)$

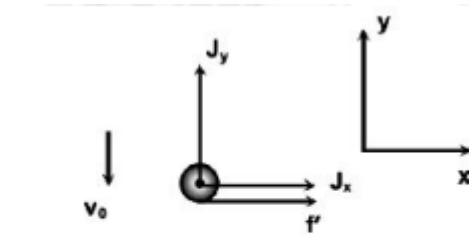
32. Circuit can be solved by using Phasor diagram.

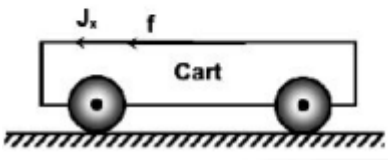
33. $\Rightarrow v_0 = \sqrt{2gh} = 5\,m/s.....(i)$

$\Rightarrow J_y = \Delta P_y = 2\times 80\times 5 = 800\ N-s.....(ii)$

$\Rightarrow J_x = \Delta P_x$

$\mu J_y = mv_x$





$v_x = 1\ m/s.....(iii)$

$\Rightarrow Mv_C = J_x$

$v_C = 0.4\ m/s.....(iv)$

$\Rightarrow t_0 = \dfrac{2\times 5}{10} = 1\ sec.....(v)$

$\Rightarrow L_{min} = 2(1+0.4)1 = 2.8\,m\ .....(iv)$

$\Rightarrow$ At the time of second collision

$R\omega' - 1 = 0.4$

$\omega' = 7\,rad/s.....(vii)$

$\Rightarrow \vec{J}' = \Delta\vec{L} - J_x R = I(\omega' - \omega_0)$

$\omega_0 = 19.5\ rad/sec\ .....(viii)$

$\Rightarrow W_{M\to m} = \dfrac{1}{2}I(\omega')^2 + \dfrac{1}{2}mv_x^2 - \dfrac{1}{2}I\omega_0^2 = -172\ J$

34. $E_2 = \dfrac{E_1 R_0}{R+R_0}\dfrac{l}{l_0}$

CHEMISTRY

35. NCERT

36. The given disaccharide has no free hemi acetal group. So it is not a reducing sugar.

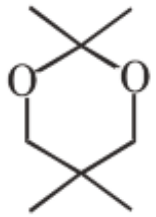
37. Cr contains 6 un paired electrons
2nd and 3rd periods contains same number of f-block elements

38. When half of the NH_4OH is neutralized a base duffer is formed.

$$p^{OH} = P^{Kb} + \log\frac{[salt]}{[Base]} \qquad [Salt] = [base] = 4.75 + \log(1) = 4.75$$

pH = 14-4.75 = 9.25

39.



40. Borax-Bead test

41. Conceptual

42. Four peroxy bonds are observed

43. $$\frac{r_{(O_3+O_2)}}{r_{O_2}} = \frac{0.4\sqrt{5}}{1} = \sqrt{\frac{MO_2}{M_{(O_3+O_2)}}} = 0.4\times0.4\times5 = \frac{32}{M_{(O_3+O_2)}}$$

$$M_{(O_3+O_2)} = \frac{32}{0.8} = 40 \text{ average} = \frac{32\times x + 48(100-x)}{100} = 40$$

$40\times100 = 32x + 4800 - 48x$

$16x = 4800 - 4000 = 800$

x = 50 and (100 – x) = 50, O_2 and O_3 are equal in percentages

$\therefore 3O_2 \rightleftharpoons 2O_3 \quad 100-3y \quad 2y$

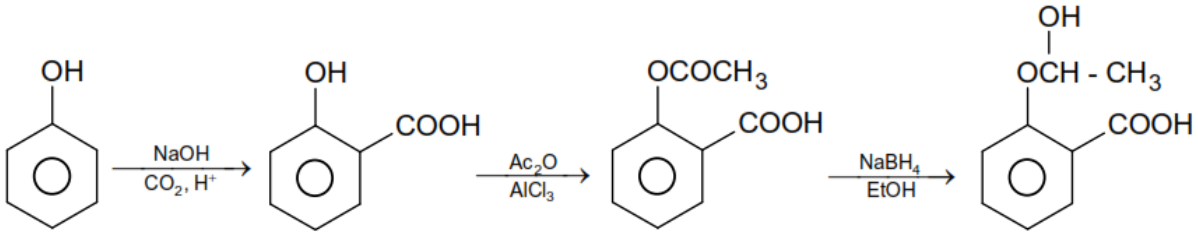
$100 - 3y = 2y \quad 5y = 100 \quad y = 20 \quad \therefore 3y = 3\times20 = 60\% =$ percentage of association

O_2 and O_3

$x\times10 = 60 \quad x = 6$

44. Product ‘P’ is benzene

45.



Aspirin

46. $K_3[Co(CN)_6]$, $K_2(Ni(CN)_4)$ contains 2 electrons in d_z^2 orbitals

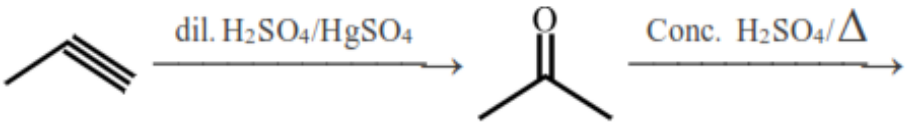
$t_{2g}^6 eg^0$

47. 1) NO mw = 30
2) NH_3 mw = 17
3) N_2 mw = 28

4) NO_2 mw = 46

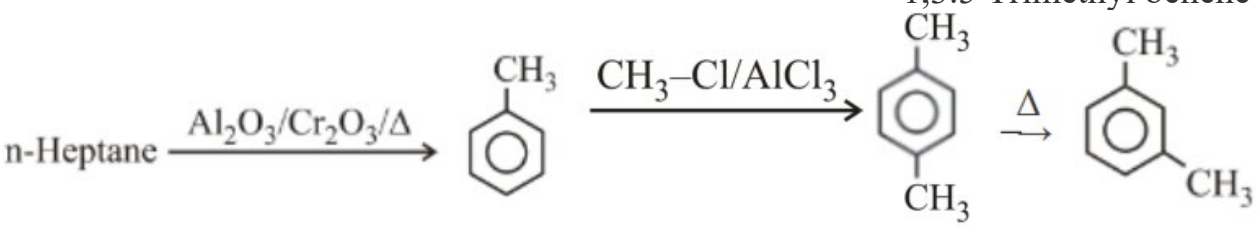
48. Conceptual

49. P)

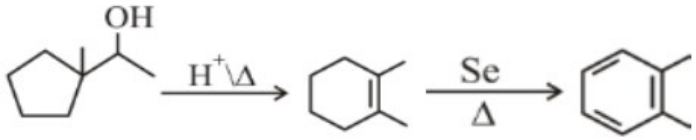


1,3.5 Trimethyl benene

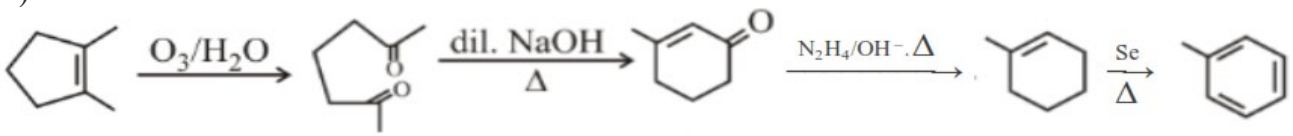
Q)



R)



S)



50. MnO_4^- colour due to $L \rightarrow MCT$

51. $2S_2F_2 + 2H_2O \rightarrow 4HF + SO_2 + 3S$